

Initialise $\hat{\alpha} = h(\overline{Y_i})$, $\hat{f}_j^0(\cdot) = 0 \forall j$, $l = 0$

for $l \leftarrow l + 1$ **do**

Construct an adjusted dependent variable Z_i according to Table 1 with

$$\hat{\eta}_i^{l-1} = \hat{\alpha} + \sum_{j=1}^p \hat{f}_j^{l-1}(X_{ij}), \quad \hat{\mu}_i^{l-1} = h^{-1}(\hat{\eta}_i^{l-1}).$$

Form the weights W_i according to Table 1.

Update the linear component (if any): fit a weighted least squares model of Z_i on the linear covariates. Update $\hat{\alpha}$ to be the fitted intercept from the parametric component, and update $\hat{f}_j^l$ for the linear terms accordingly.

Update the non-linear component: fit a weighted additive model to Z_i using the backfitting algorithm (Algorithm 2) with weights W_i to obtain the estimated functions $\hat{f}_j^l$, additive predictor $\hat{\eta}_i^l$, and fitted values $\hat{\mu}_i^l$.

Compute the convergence criterion $\text{DEV} < \delta$, with DEV as in Table 1.

Until the convergence criterion is satisfied